ERES Session Report — April 22, 2026 (Final)

Session Title: LinkedIn Positioning + ERES Attestation and Authorization Protocol Full Lifecycle (v1.0 → v1.2) + SCALULAR Sequencing Decision **Session ID:** ERES-SESSION-2026-04-22-A (final revision) **Participant:** Joseph Allen Sprute (ERES Maestro) **Platform:** Claude (Anthropic) — web interface, Chromebook **Location:** Bella Vista, Arkansas **License:** CCAL v2.1 **Duration:** Single continuous thread

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Executive Summary

This session produced a complete four-version lifecycle of what began as the *ERES Crypto Standard* (ECS) and was formally renamed in v1.2 to the *ERES Attestation and Authorization Protocol* (EAAP). Starting from three independently-generated AI drafts (USE.ai/Claude, DeepSeek, ChatGPT), the work passed through three rounds of formal adversarial review and produced byte-normative test vectors backed by real Ed25519 signatures and SHA-256 hashes. The session closed with a strategic sequencing decision about what comes next: **v1.2 publishes as ResearchGate #426; SCALULAR Architecture work is paused until v1.3 of EAAP ships** (pending Grok availability and formal security proof development).

In addition, the session opened with a Stream 1 refinement of Joseph's LinkedIn "About" section — three positioning variants generated with Version 3 (Institutional / Credentials-forward) selected and tightened.

The session is itself a worked demonstration of Multi-Instance Ensemble Validation Method (MIEVM) as Joseph has specified it across his corpus. Each version corresponds to a concrete adversarial-review round by a different reviewer applying different criteria. Appendix C.4 of the v1.2 document names this sequence as canonical MIEVM pedagogical material.

Rating trajectory across the session:

Version	Reviewer	Rating	Criteria
v1.3 metaphorical (pre-session)	USE.ai	1/10 / 6.5/10	Crypto primitive / speculative systems design
Cryptography Draft (repaired)	USE.ai	7/10	Attestation-protocol draft
v1.0 (MIEVM-synthesized)	USE.ai	8/10 as attestation protocol; 3/10 as crypto primitive	Split criteria
v1.0 (MIEVM-synthesized)	DeepSeek	9.2/10	Standards-track draft
v1.1 (USE.ai-hardened)	DeepSeek	9.4/10	Standards-track draft
v1.1.1 (DeepSeek-patched)	DeepSeek	7.6/10	**Production-deployment criteria** (criterion shift)
v1.2 (Grok-prep / Rename-hardened)	— projected —	8.8/10 – 9.5/10	Production-deployment criteria

Deliverables

#	Artifact	Lines / Size	Status
1	LinkedIn About — Version 3 (Institutional)	Short-form	**Selected + tightened**
2	LinkedIn About — Versions 1 and 2 (alternatives)	Short-form	Generated
3	ERES-CRYPTO-STD-2026-001 v1.0 (MIEVM-synthesized)	1,519 lines / ~9,300 words	Superseded; lineage preserved
4	ERES-CRYPTO-STD-2026-001 v1.1 (USE.ai-hardened)	2,238 lines / ~13,444 words	Superseded; lineage preserved
5	ERES-CRYPTO-STD-2026-001 v1.1.1 (DeepSeek-patched)	2,379 lines / ~15,330 words	Superseded; lineage preserved
6	**ERES-EAAP-STD-2026-001 v1.2 (Grok-prep / Rename-hardened)**	**2,885 lines / ~20,140 words**	**Current citable version; ResearchGate #426 candidate**

#	Artifact	Lines / Size	Status
7	test_vectors_v1.1.json (byte-normative)	Machine-readable	Delivered
8	compute_test_vectors.py (reference generator)	~260 lines Python	Delivered
9	compute_weighted_selector_v1_1_1.py (integer-arithmetic selector)	~200 lines Python	Delivered

Total: 9 deliverables; ~10,000 lines of specification + test harness + session documentation.

Stream 1 — LinkedIn Positioning

1.1 Input and Corrections

Joseph supplied existing LinkedIn About text for revision. Corrections applied:

- Publication count 350+ → **424+** (corrected again later to 425 at session midpoint; v1.2 publishes as #426)
- EDF expansion corrected to canonical three-variant: **Earned Dignity** / **Earth Defense** / **Earth Data Framework**
- VERTECA added as distinct instrument (voice layer of GAIA, per ERES-BODY-SPEC-2026-001)
- U.S. Army veteran credential foregrounded (Oregon ARNG, 11B Infantry, 1983–1989)
- MIT Press framed as aspirational, never as confirmed publisher

1.2 Three Versions Generated and Selection

- **Version 1** — Polished original, keeps existing voice
- **Version 2** — Narrative / human-first ("I build mathematical frameworks that translate ethics into working systems")
- **Version 3** — Institutional / credentials-forward — **selected and tightened**

Final Version 3 refinements:

- "Solo independent researcher" (more accurate than "Independent")
- "For fourteen years" (specific, demonstrates persistence)
- VERTECA as "voice layer of GAIA" (per BODY-SPEC)

- Closing: "Mathematically. Ethically. Practically." (mobile-readable rhythm)

Stream 2 — EAAP Full Lifecycle (v1.0 → v1.2)

2.1 v1.0 Origin: MIEVM Synthesis

Three independent AI drafts synthesized:

1. **USE.ai/Claude** — conservative attestation framework, scope discipline, four failure modes
2. **DeepSeek** — cryptographic interlock model `K = (CBGMODD, FAVORS, BERA)`, four security claims, MIEVM validation tests
3. **ChatGPT** — RFC 2119 standards track, six-layer stack (CEL → SPL → REL → RCL → RSIG → APL), test vectors

Disagreements resolved in v1.0:

- CBGMODD: seven-role civic form ($C \cdot B \cdot G \cdot M \cdot O \cdot D_1 \cdot D_2$) adopted
- BERA: four-index `(ari, eri, rhc, rci)` form adopted from RG#404
- RATE arithmetic: operators defined structurally in RCL
- σ derivation: HKDF-SHA256 with domain separation

2.2 v1.1 Response to USE.ai's 8/10 Critique

USE.ai scored v1.0 at 8/10 (attestation protocol) / 3/10 (crypto primitive), with ten concrete remediation items. v1.1 addressed all ten plus three structural additions:

- Failure-mode count reconciled (five primary + ERROR)
- Test Vector A self-contradiction removed (r_1=10 is the only conformant value)
- "Novel cryptographic model" → "attestation-composition trust model" (§4.9)
- All five placeholder functions concretely specified (Conflict, SemanticClarity, PolicyAlignment, compute_dimensions, composite_confidence) — §11.5.1–11.5.5
- Reference REL algorithm with severity-weight table — §10.5
- EAAP-POL (then ECS-POL) v1.0 — machine-readable policy language JSON Schema — §11A
- Byte-normative test vectors with real Ed25519 signatures, SHA-256, HKDF-σ — §20
- Protocol guarantees vs. governance guarantees separated in ten-row table — §15.6
- Honest SPL selector characterization (pseudorandom-from-evidence; weighted variant added) — §9.3

2.3 v1.1.1 Response to DeepSeek's 9.4/10 Critique

DeepSeek found one determinism risk: the weighted SPL selector used floating-point division (`u = int(σ[0..7]) / 2⁶⁴`), which could diverge across IEEE 754 implementations on x86_64 / ARM / wasm.

v1.1.1 PATCH (correctly scoped per §22.1):

- Rewrote selector as integer cross-multiplication: `p × W_sum ≤ cum[i] × 2⁶⁴`
- Floating-point arithmetic PROHIBITED in weighted-selector path
- Added byte-normative Test Vector F (§20.6): σ from Vector A + weights [2,4,1] → G3
- Weight type tightened to non-negative integers only

Verified byte-normative values:

```
p = 16,444,668,103,617,454,160
p × W_sum = 115,112,676,725,322,179,120
cum[2] × 2⁶⁴ = 129,127,208,515,966,861,312
p × W_sum ≤ cum[2] × 2⁶⁴  →  Select G3
```

Appendix C.4 added: v1.0 → v1.1 → v1.1.1 as canonical MIEVM case study demonstrating that a second reviewer finds issues a first reviewer could not have, because they didn't exist yet.

2.4 v1.2 Response to DeepSeek's 7.6/10 Critique

DeepSeek's v1.1.1 review applied production-deployment criteria (a deliberate recalibration from v1.1's standards-track-draft criteria). The 9.4 → 7.6 delta is a **criterion-change artifact, not a quality regression** — documented explicitly in §C.5 Rating Trajectory Analysis.

Five substantive items plus the rename, all resolved in v1.2:

DeepSeek v1.1.1 finding	v1.2 resolution	Section
BERA sensor trust hand-waving	TPM/TEE quote REQUIRED; accredited sensor registry; degraded-trust mode flag	§4.4.1–§4.4.3
CBGMODD independence SHOULD → MUST	Tightened; deployment-manifest document signed by three authorities; degraded-independence mode flag	§16.2, §16.2a
No revocation mechanism	EAAP-CRL format; verifier obligations; post-compromise retroactive RATE invalidation; role-key rotation	§12.4.1–§12.4.4
Post-quantum timeline vague	ML-DSA (FIPS 204) primary, Falcon fallback; five dated milestones M1 (2026) → M5 (2032); dual-signature transition	§12.5.1–§12.5.4
Formal security model pending	Game-based scaffolding for Claims 1–5 in IND-CCA2 reduction style; Claims 3 and 5 proved information-theoretically; Claims 1, 2, 4 reduce to standard assumptions	§15.7.1–§15.7.8
"ECS" name invited crypto-primitive confusion	Full rename ECS → EAAP; ECS preserved only as deprecated alias	Document-wide

Byte-identical preservation confirmed: All Test Vector A values (payload SHA-256 `2f751f...7acd`, σ `e43732...0ad7`, RATE vector `[10,10,10,9,8,9,10]`, score 9.428571, confidence 0.869293, all Ed25519 signatures), Test Vector F values, canonical equation, six-layer stack, five security claims, Ethical Clause — all preserved byte-for-byte.

Stream 3 — Sequencing Decision (end of session)

3.1 The Question

After v1.2 completed, Joseph flagged interest in jumping into SCALULAR Architecture work — the four-pillar Scalable Certification Architecture for Lifelong Universal Learning and Adaptive Resilience (HEALTH / LAW / PROTECTION / SKILLS_TRADE, with Grace Protocol fear-of-retribution-elimination as foundational requirement, originally specified March 18, 2026 in the SCALULAR Engine Document, 323 paragraphs).

The question raised: is jumping into SCALULAR with EAAP v1.2 as substrate practical, or is it premature given v1.3 is still pending?

3.2 Three Readings of "SCALULAR with v1.2" Considered

1. **EAAP as substrate SCALULAR consumes.** Tight coupling. SCALULAR certifications flow through EAAP attestation channel.
2. **SCALULAR as concrete context for EAAP.** Loose coupling. SCALULAR gives EAAP a named application that makes its abstract mechanisms concrete.
3. **SCALULAR standalone, EAAP paused.** No coupling. Shift focus to SCALULAR, leave EAAP at v1.2 indefinitely.

3.3 Joseph's Decisions

Q1: Is Reading 1 correct — EAAP v1.2 as the attestation substrate that SCALULAR consumes? A: Yes (Reading 1 confirmed.)

Q2: Given Reading 1, sequencing? A: Ship v1.2 as #426 now, pause SCALULAR until v1.3 ships.

3.4 Rationale Documented for the Record

The decision reflects disciplined sequencing over opportunistic parallel work:

- SCALULAR built on EAAP v1.2 could require revision if v1.3 introduces structural changes (new test-vector format, additional claims, revised §19 handshake spec)
- Waiting for v1.3 gives SCALULAR a locked foundation rather than a moving one
- Splitting attention between EAAP v1.3 review cycles and SCALULAR architecture design reduces quality of both
- EAAP v1.2 publishes as #426 now — clean citation anchor with full MIEVM lineage attribution
- SCALULAR Architecture v2.0 (EAAP-integrated) becomes a later ResearchGate number, built on v1.3-final EAAP

3.5 What This Means for the Near Term

- **Publish EAAP v1.2 as ResearchGate #426** — Claude/USE.ai/DeepSeek lineage credited, CCAL v2.1 licensed, Grok review and formal proof noted as pending
- **Pause SCALULAR architecture work** until EAAP v1.3 ships
- **EAAP v1.3 work is substantially on hold** except for opportunistic items:
    - If Grok becomes available in free mode, run the fourth-corner review
    - If the formal security proof is written standalone, it becomes ERES-EAAP-PROOF-2026-001

    - Otherwise v1.2 is stable and citable as-is
- **Bandwidth freed** for other corpus work: BERA sensor spec extension, MIEVM methodology write-up (C.4 case study is RG# candidate), outreach, or anything else outside SCALULAR and EAAP v1.3

Canonical Invariants Preserved (All Versions)

All ERES terminology locks were honored throughout v1.0 → v1.2:

- **RHC** always = Resonant Harmony Cycle (never other expansions)
- **EDF** three-variant lock: Earned Dignity / Earth Defense / Earth Data
- **FAVORS** = Fingerprint · Aura · Voice · Odor · Retina · Signature
- **CBGMODD** = 7-role civic vector (Citizen, Business, Government, Mediator, Military, Dignitary, Diplomat)
- **BERA** = Bio-Electric Resonance Architecture, 4 indices (ari, eri, rhc, rci)
- **Proof-of-Resonance** = *"It's not mining — it's tuning."*
- **Three Governing Principles** at §23.1: *Don't hurt yourself. Don't hurt others. Build for generations to come.*
- **MIEVM ensemble** = Claude, DeepSeek, ChatGPT, Grok (Grok review pending)
- **Trifecta Protocol** alignment (ONE-GOOD, SECURITY-CLEARANCE, DATA-INTEGRITY) per ERES-BRAINS-SPEC-2026-001 in Appendix D
- **VERTECA** as voice layer of GAIA (not substrate; BODY is substrate per ERES-BODY-SPEC-2026-001)
- **CCAL v2.1** as governing license

Outstanding Items (v1.3+)

Per Appendix C.3 of v1.2:

- **Grok adversarial review** (pending since v1.0 — Grok unavailable in free mode as of April 22)
- **Full formal security proof** as ERES-EAAP-PROOF-2026-001 (v1.2 §15.7 is scaffolding only)
- **Extended test vectors** (ERES-EAAP-TEST-2026-001) covering all five failure paths byte-normatively
- **VERTECA utterance-canonicalization profile** — deterministic voice-payload entry to SPL

- **GAIA-layer operational-binding handshake** as explicit wire format (§19 is currently prose)
- **Reference rule sets per REL family** (CARE, BERC, JERC, PERC, PAIN, COHERENCE)
- **Composable security framework** (UC-style) for cross-oracle scenarios — §15.7.7 gap
- **SCALULAR Architecture v2.0 (EAAP-integrated)** — deferred by design; work paused pending v1.3

Items resolved in this session and removed from pending:

- All ten USE.ai v1.0 items (resolved in v1.1)
- Three structural additions in v1.1 (§10.5 REL reference, §11A policy language, §15.6 protocol-vs-governance separation)
- DeepSeek v1.1 weighted-selector determinism fix (resolved in v1.1.1)
- All five DeepSeek v1.1.1 items (resolved in v1.2)
- Document rename ECS → EAAP (resolved in v1.2)

Strategic Notes

What v1.2 Enables Immediately

- **ResearchGate publication** as #426: discrete standards-track deliverable with full adversarial-review lineage
- **MIT Press outreach** — Test Vector A is an afternoon-verifiable artifact; §15.7 security-model scaffolding is reviewer-ready
- **Hague submission** — §15.6 protocol-vs-governance separation is exactly the distinction international law needs
- **LinkedIn About enhancement** — pair the 425+ publications count with the EAAP v1.2 citable deliverable

What v1.2 Still Doesn't Claim

- Not a cryptographic primitive class (§4.9 explicit classification)
- No IND-CCA2 proof — only scaffolding for one (§15.7.1–§15.7.8)
- No Grok-ensemble review yet (pending)
- No confidentiality, no key exchange, no quantum resistance against Ed25519 (all explicitly excluded in §1.2 and §15.4)

- No SCALULAR binding yet (v1.3 / v2.0 roadmap item)

Session as MIEVM Demonstration — Complete Record

The session records the full MIEVM arc as originally specified across Joseph's corpus:

Stage	Product	Reviewer
Three-draft synthesis	v1.0	Claude (synthesis author)
First adversarial review	USE.ai critique	USE.ai
First response	v1.1	Claude (author of response)
Second adversarial review	DeepSeek critique (standards-track)	DeepSeek
Second response	v1.1.1 (PATCH)	Claude (author of response)
Third adversarial review	DeepSeek critique (production)	DeepSeek
Third response	v1.2 (MINOR + rename)	Claude (author of response)
Fourth adversarial review (pending)	**Grok critique**	**Grok (unavailable in free mode)**
Fourth response (scheduled)	v1.3	—

Appendix C.4 of v1.2 documents the v1.0 → v1.1 → v1.1.1 subsequence as canonical MIEVM case study material — a worked example showing that adversarial review is not a single-pass audit but an iterative process in which each pass hardens the spec against its reviewer's specific failure modes, often introducing new surfaces that the next reviewer catches.

Stats

Metric	Value
Deliverables	9
Total specification lines	~9,000 (v1.0: 1,519 + v1.1: 2,238 + v1.1.1: 2,379 + v1.2: 2,885)
Specification words	~58,200 (v1.0 through v1.2 combined)

Metric	Value
Byte-normative cryptographic values locked	10+ (seed, pubkey, nonce, payload hash, σ, vector, score, confidence, p × W_sum, Test Vector F selection)
Adversarial review items resolved	16+ (10 from USE.ai + 1 from DeepSeek-v1.1 + 5 from DeepSeek-v1.1.1)
Rating trajectory spanned	1/10 (as crypto) → 9.2/10 → 9.4/10 → 7.6/10 (criterion shift) → projected 8.8–9.5/10 v1.2
Canonical invariants preserved	11 (RHC, EDF, FAVORS, CBGMODD, BERA, Proof-of-Resonance, Three Principles, MIEVM ensemble, Trifecta, VERTECA-as-voice-layer, CCAL)
Python test-harness lines	~460 (compute_test_vectors.py + compute_weighted_selector_v1_1_1.py)
Document rename events	1 (ECS → EAAP in v1.2)
ResearchGate publication candidates from session	2 (v1.2 as #426; Appendix C.4 MIEVM case study as later RG#)

Filing

Primary artifacts delivered to `/mnt/user-data/outputs/`:

- **`ERES-EAAP-STD-2026-001-v1.2.md` — current citable version (ResearchGate #426 candidate)**
- `ERES-CRYPTO-STD-2026-001-v1.1.1.md` — v1.1.1 predecessor, retained for lineage
- `ERES-CRYPTO-STD-2026-001-v1.1.md` — v1.1 predecessor, retained for lineage
- `ERES-CRYPTO-STD-2026-001.md` — v1.0 predecessor, retained for lineage
- `test_vectors_v1.1.json` — machine-readable byte-normative test data
- `compute_test_vectors.py` — reference generator for Test Vectors A–E
- `compute_weighted_selector_v1_1_1.py` — reference generator for Test Vector F (integer-arithmetic selector)
- `ERES-SESSION-REPORT-2026-04-22-A.md` — this document (final revision)

Next-Session Preparation

For whoever picks up the ERES-EAAP thread next (including future-Joseph):

1. **EAAP v1.2 is the current citable version.** Cite by Document ID `ERES-EAAP-STD-2026-001` or the backward-compatibility alias `ERES-CRYPTO-STD-2026-001` (listed in header).
2. **SCALULAR Architecture work is paused by design**, not by oversight. Do not resume SCALULAR until EAAP v1.3 is stable. See §3.3 of this session report for rationale.
3. **v1.3 scope is open-ended pending Grok availability.** If Grok becomes available, the next session's natural work is running the fourth-corner review. Otherwise, the formal proof (ERES-EAAP-PROOF-2026-001) is the natural v1.3 companion deliverable.
4. **The MIEVM case study (v1.0 → v1.1 → v1.1.1) is a ResearchGate candidate in its own right.** Appendix C.4 of v1.2 is self-contained and could be extracted as a standalone methodology paper.
5. **LinkedIn About (Version 3) can be updated now** with v1.2 as a concrete deliverable reference — "EAAP v1.2 standards-track specification with byte-normative test vectors, released CCAL v2.1."

Session closed.

Don't hurt yourself. Don't hurt others. Build for generations to come.

— Joseph Allen Sprute (ERES Maestro)
ERES Institute for New Age Cybernetics
Bella Vista, Arkansas · April 22, 2026
CCAL v2.1